

Using DECT NR+

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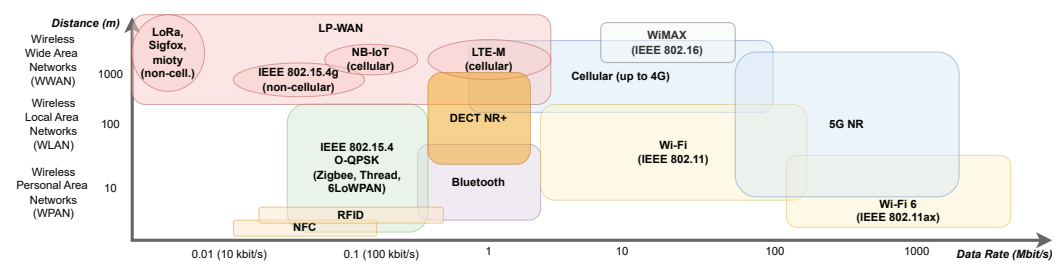
2026-09-01
1AD, Grenoble

Plan for today

1. Briefly: Introducing DECT NR+
2. Hardware
3. Software
4. What we can do with it

What is DECT NR+...

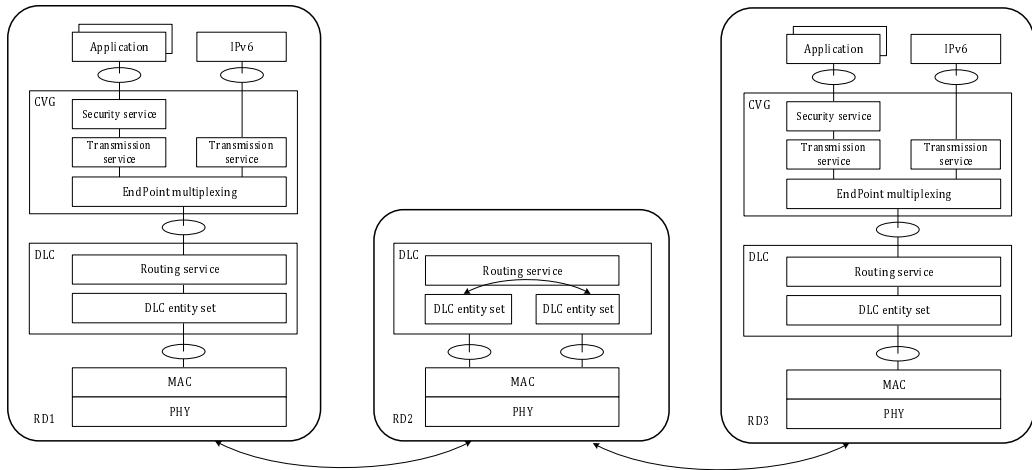
...in the context to radio technologies you know?



¹F. Graf, T. Watteyne and M. Villnow, "What to Expect When Using DECT NR+", in IEEE Communications Magazine, vol. 64, no. 2, pp. 16-22, February 2026, doi: [10.1109/MCOM.001.2500286](https://doi.org/10.1109/MCOM.001.2500286)

Radio stack

271 pages in 1 picture



²Diagrams from ETSI TS 103 636-[1-5] v2.1.1 unless noted otherwise.

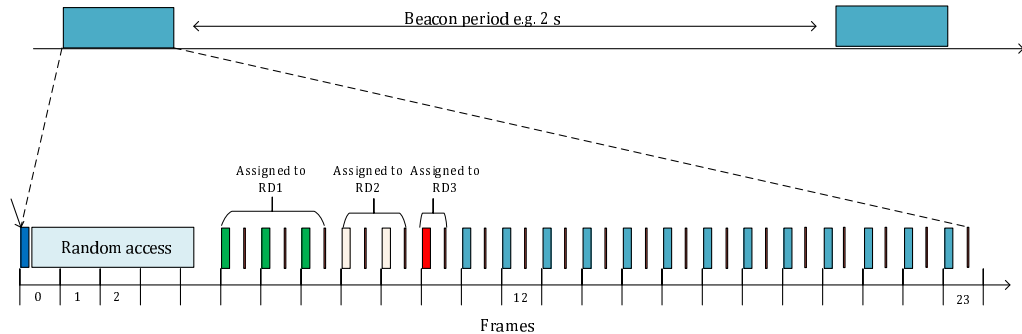
PHY layer: key properties

- ▶ 20 MHz at 1.9 GHz (new bands incoming)
- ▶ Error correction tools in PHY: FEC, HARQ, CRC
- ▶ OFDM, modulation
- ▶ Throughput depends on SNR
- ▶ Buffers limit packet size

MAC: Roles & topology

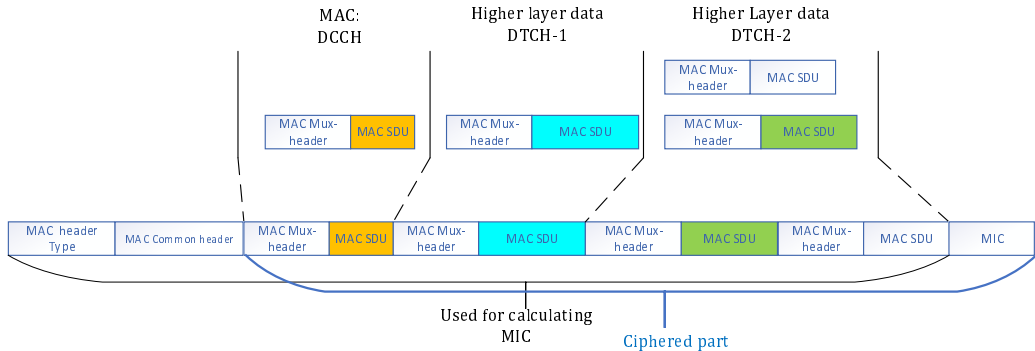
- ▶ Cluster: coordinated by “Fixed Termination points” (FT) associated with “Portable Termination points” (PT)
- ▶ Tree topology. Or just star. Or point-to-point.
- ▶ Root: “Sink”

MAC: Cluster Beacons



MAC

Data in PDC (PHY data channel)



MAC: Data

Typical elements found

- ▶ Beacon details
- ▶ Association management

Forms tree, or star, or point-to-point connection.

- ▶ Data item(s!)
- ▶ Security: AES. Keys “not our department” – CoJP/LAKE to the rescue?

IP over DECT NR+

- ▶ Full IPv6 packet or 6lo header compression
- ▶ Routed based on local part constructed from DECT addresses



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SiP 12x11 mm

nRF9151

System-in-Package

Low power SiP with integrated LTE-M/
NB-IoT/NTN and DECT NR+ modem,
and GNSS

The nRF9151 module sets a new standard for highly integrated and compact System-in-Package (SiP) solutions, specifically designed for [cellular IoT](#) and [DECT NR+](#) applications. Leveraging low power LTE technology, advanced processing capabilities, and robust security features, the nRF9151 offers unparalleled performance and versatility, and supports 3GPP release 14 LTE-M/NB-IoT, NB-NTN (Non-Terrestrial Network) and DECT NR+.

Compared to its predecessor ([nRF9160](#)), it boasts a significant footprint reduction of 20% and brings additional support for Power Class 5 20 dBm.

Where to Buy

NRF9151-LACA-R ▾

Newark
3866 in stock
Buy Now

Mouser Electronics

Key features

- Fully integrated SiP module for cellular IoT, NTN and DECT NR+
- Multimode LTE-M/NB-IoT/NTN modem with DECT NR+ support and GNSS
- 700-2200 MHz LTE band support
- Power Class 5 20 dBm
- Power Class 3 23 dBm
- 915 MHz & 1.9 GHz NR+ band support

Hardware

<https://www.lm-semi.com/lm10xx-series-system-on-chip/>



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LM10xx Series

Designed for next-generation IoT Connectivity.

⊕ Ultra-Secure ⊕ Highly reliable ⊕ Low-cost ⊕ Simple to implement



Our LM10xx Series System-on-Chip is based on the globally free, non-cellular 5G standard NR+, provides a low-cost, secure and reliable

Available hardware: nRF9151

- ▶ Application core
- ▶ Locked network core
- ▶ Default firmware: LTE
- ▶ Alternative firmware: NR+ PHY and MAC

Implementations on nRF9151

- ▶ Proprietary
 - ▶ Wirepas – metering
 - ▶ Lynq – audio
 - ▶ Stratum9 – industrial / deterministic
 - ▶ DSR – IoT
- ▶ Research: MERCI project
- ▶ Working examples: Nordic's DECT shell (on MAC firmware)



Opener Initiative

- ▶ full implementation, eventually

- ▶ building bottom-up

 - current work on drivers and MAC layer

- ▶ developed on Zephyr, written in C



hophop

- ▶ making DECT available in Rust
- ▶ focus: on high-level

IPv6 for Ariel OS

- ▶ currently on nRF MAC; plan to switch to Opener
- ▶ contributing to Opener

Using hophop today: Documentation

<https://github.com/ariel-os/hophop/>

Getting started with hophop

Basic steps common to all examples and applications

- Get two (better four) [nRF9151-DK](#), connect them and turn them on.

(Wider hardware support will be added, but this is what the examples assume).

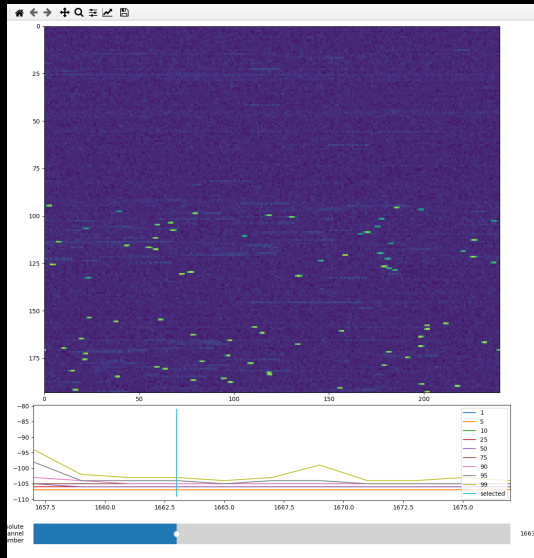
- Ensure you are set up to run Ariel OS examples.

For the time being, it's easiest to follow the [Getting Started section in the Ariel OS book](#).

Note that in particular, you need to have access to the boards you are using as a user; this is only mentioned in the Ariel documentation indirectly by referring to the [probe-rs documentation](#). That includes verifying that your user is part of the `plugdev` group, and placing [the probe-rs rules file](#) inside `/etc/udev/rules.d`. The typical error that indicates that this step is missing is "USB device: permission denied".

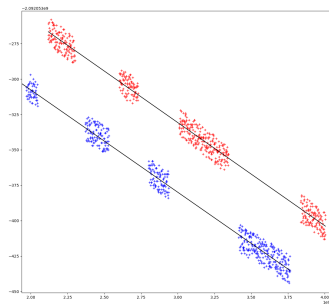
- [Ensure that you have the DECT firmware running](#).
- Beware that this is a research example, and that depending on your location, regulation on operating these devices does apply, especially as some examples emit transmissions controlled by the user alone.

Using hophop today: RSSI (on PHY)

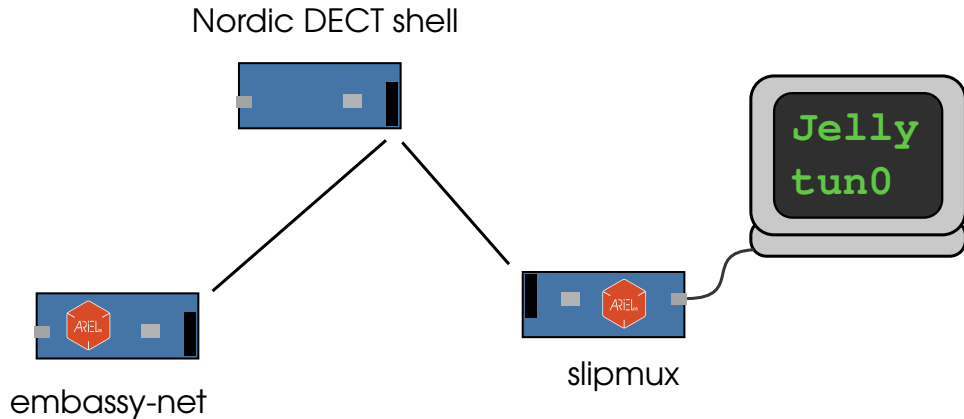


Using hopop today: more PHY examples

- ▶ Transmit
- ▶ Receive and decode
- ▶ Measure clock drift



Using hopsho today: IP transport



Using hop hop today: IP transport

```
$ ping fe80::70d1:776d:f9cf:c820%tun0  
64 bytes from ...: icmp_seq=1 ttl=64 time=3077 ms  
64 bytes from ...: icmp_seq=2 ttl=64 time=2086 ms
```

Using hophop today: IP transport

```
$ ping fe80::70d1:776d:f9cf:c820%tun0
```

```
64 bytes from ...: icmp_seq=1 ttl=64 time=3077 ms
```

```
64 bytes from ...: icmp_seq=2 ttl=64 time=2086 ms
```

```
$ aiocoap-client 'coap://[fe80:...]/hello' --credentials cred.diag  
Hello from hophop
```

Next steps

- ▶ FT role
- ▶ Automatic network configuration
- ▶ Star topology: Opener
- ▶ Telemetry and remote configuration
- ▶ Secure joining: ETSI?

Run an IoT network.



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